Budget Front Desk

An agent built from public business pages, measured before and after

2026-11-10

The deliverable

A live Luna front desk for Budget Heating, Cooling & Plumbing, built from a **24-field v1 brief** with page-and-line provenance. The current ledger has 25 fields after adding the omitted membership benefits; v2 has not been evaluated. The agent triages requests, uses the catalogue, collects intake and routes to a booking page or phone. All large test sets and recorded runs are retained.

Before and after

	A0 baseline	A5 instructions
single-turn pass@1, respective 53/55 scenarios	113/159 (71.1%)	145/165 (87.9%)
strict pass[3], shared 53 scenarios	26/53	**43/53**
single-turn integrity-gate failures	0/53	0/55
multi-turn, 54 conversations	36/54 (67%)	**41/54 (76%)**
multi-turn scripts passing all trials	9/18	**12/18**

The paired strict-pass result favours A5 on the development set (b=18, c=1, p<0.001). Pass@1 uses different denominators and is descriptive. The original single-turn metadata names project `jojuduewx7ud`; the linked A5 rebuild, `cgiefgaebmgh`, has its own multi-turn and sealed measurements. The multi-turn baseline now exists: A0 is 36/54 and A5 is 41/54. A5 gains 7 conversations and loses 2; McNemar p=0.180. The measured direction is positive, but the multi-turn delta is not statistically settled. These are three trials of 18 scripts, not 54 independent customers; one gain is A0's unfilled-placeholder integrity failure, so the clean behavioural signal is narrower than 7 versus 2.

What changed

A1's generic route instruction permitted "reply in chat". A2 required an external destination. A3's hours instruction prompted a clock-based contradiction; A5 kept the customer's stated time and forbade clock checks. The arm history retains the regressions as well as the selected changes.

The largest multi-turn gains are **arithmetic (2/9 to 5/9)** and **pushback (5/9 to 7/9)**. Memory stays 9/9 in both arms; recovery stays 9/9; change-of-mind stays 3/9. One A0 failure emitted an unfilled template placeholder; A5 passed that conversation.

The check I could not tune against

Twenty three-turn scenarios were authored by a different model, with graders frozen before replies existed. On the original run A0 and A5 both scored **17/20**, agreeing on 18/20; on the 13 unambiguous scenarios A0 scored 12/13 and A5 11/13. One A5 reply was then excluded by deterministic cleaning because a render failure was appended to a complete answer; excluding it leaves A5 at 16/19 overall and 10/12 on the scored scenarios. The result does not confirm the single-turn gain out of sample, and the set is too small to call it a regression.

Limits and failures

- The single-turn comparison is in-sample. There are **21 logged corrections**: 20 from transcript inspection and one from a positive control. Corrections 20 and 21 repaired amount checks that rejected valid arithmetic, including $99 + $150 + $50 = $299. No frozen scenario rubric was changed.
- Six failing A5 conversations include savings-club expectations absent from the brief; some also miss a required diagnostic price. The overnight example in MT-P02 declines service but fails a wording-sensitive regex. Scores are unchanged.
- One build showed website-accurate savings-club benefits that were absent from the brief while another build declined. The cause is unknown; the repository reports it as an open platform question, not as a proven defect.
- A human calibration reached κ=0.911 only after the rule was fitted to the same 30 transcripts; it is not fresh-sample agreement.

The complete evidence is in `RESULTS.md`, `RIGOR.md`, `FAILURES.md` and `FINDINGS-FOR-AGENTPLACE.md`. The live agent link is supplied separately.